Treatment of the Coulomb Interaction in Solids

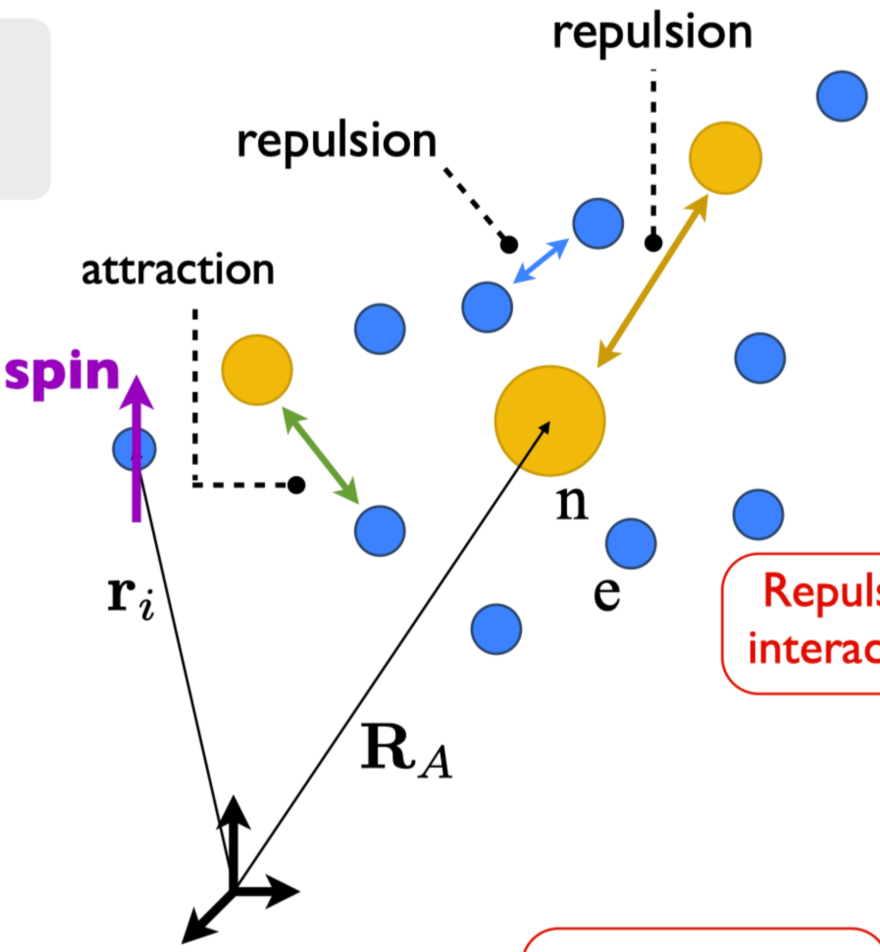
Jacques K. Desmarais

- Hamiltonian
- Energy functional
- Cauchy integral test of convergence
- Ewald summation technique
- Saunders et al. point multipole model
- References

The Hamiltonian

THE MANY-ELECTRON PROBLEM

e = electron
n = nucleus



Atomic units: $e = m = \hbar = 1$

$$\hat{H} = \hat{T}_{\mathrm{e}} + \hat{V}_{\mathrm{ee}} + \hat{V}_{\mathrm{ne}} + \hat{V}_{\mathrm{nn}}$$

$$\hat{T}_{\mathrm{e}} = -\sum_i \frac{\nabla_i^2}{2}$$

Kinetic operator

Repulsive interaction

$$\hat{V}_{\mathrm{ee}} = \frac{1}{2} \sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

$$\hat{V}_{\mathrm{en}} = \sum_i \sum_A \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|}$$

External potential

A constant (for fixed nuclei)

$$\hat{V}_{\mathrm{nn}} = \frac{1}{2} \sum_{A \neq B} \frac{Z_A Z_B}{|\mathbf{R}_A - \mathbf{R}_B|}$$

The Energy Functional

$$F[n] = \langle \Psi | \hat{T} + \hat{V}_{\text{ee}} | \Psi \rangle := \boxed{T_{\text{KS}}[n]} + \boxed{\int_{\text{cell}} d^3r \int_{\text{cell}} d^3r' \sum_{\mathbf{R}} \frac{n(\mathbf{r})n(\mathbf{r}')}{2|\mathbf{r} - \mathbf{r}' + \mathbf{R}|}} + \boxed{E_{\text{xc}}[n]}$$

$$E = F[n] + \int_{\text{cell}} d^3r \, n(\mathbf{r}) v_{\text{en}} + E_{\text{nn}} = \boxed{T_{\text{KS}}[n]} + \boxed{E_{\text{xc}}[n]} + \boxed{\int_{\text{cell}} d^3r \int_{\text{cell}} d^3r' \sum_{\mathbf{R}} \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{2|\mathbf{r} - \mathbf{r}' + \mathbf{R}|}}$$

$$\rho(\mathbf{r}) = \sum_A Z_A \delta(\mathbf{r} - \mathbf{r}_A) + n(\mathbf{r})$$

Total charge

Charge neutrality condition

$$\int_{\text{cell}} d^3r \, \rho(\mathbf{r}) = 0$$

$$E_{\mathrm{Coul}} = \int_{\mathrm{cell}} d^3r \int_{\mathrm{cell}} d^3r' \sum_{\mathbf{R}} \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{2|\mathbf{r} - \mathbf{r}' + \mathbf{R}|} = \frac{1}{2} \int_{\mathrm{cell}} d^3r \; \rho(\mathbf{r})\Phi[\rho](\mathbf{r})$$

$$\Phi[\rho](\mathbf{r}) = \int_{\mathrm{cell}} d^3r' \sum_{\mathbf{R}} \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}' + \mathbf{R}|}$$

Does the series converge?

Convergence properties of lattice series

- Consider the infinite series

$$\sum_{R=A}^{\infty} f(R)$$

- According to the *Cauchy integral test*, the series is convergent if and only if the improper integral

$$\lim_{B\to\infty} \int_A^B dr \; f(r) r^{d-1}$$

- has a finite value. If the integral diverges, then the series can be conditionally convergent or divergent.

$$f(R) = \frac{1}{R} \qquad d = 1$$

$$\sum_{R=1}^{\infty} \frac{1}{R^{1+\varepsilon}} \quad \text{with} \quad \varepsilon > 0$$

$$\sum_{R=1}^{\infty} \frac{1}{R}$$

$$\lim_{B\to\infty} \int_1^B dr\, \frac{1}{r^{1+\varepsilon}} = \frac{1}{\varepsilon}$$

$$\lim_{B\to\infty} \int_1^B dr\, \frac{1}{r} = \lim_{B\to\infty} ln(r)\big|_1^B = \lim_{B\to\infty} ln(B)$$

$$\Phi[\rho](\mathbf{r}) = \int_{\text{cell}} d^3r' \sum_{\mathbf{R}} \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}' + \mathbf{R}|} = \sum_{\mathbf{R}} \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \eta_\ell^m[\rho](\mathbf{C}) Z_\ell^m(\partial/\partial\mathbf{C}) \frac{1}{|\mathbf{r} + \mathbf{R} - \mathbf{C}|}$$

$$Z_\ell^m(\partial/\partial\mathbf{C}) = \frac{(2 - \delta_{m0})[(\ell - m)!]}{[(2\ell - 1)!!][(\ell + m)!]} X_\ell^m(\partial/\partial\mathbf{C})$$

$$\eta_\ell^m[\rho](\mathbf{C}) = \int_{\text{cell}} d^3r' \; \rho(\mathbf{r}') X_\ell^m(\mathbf{r}' - \mathbf{C})$$

$$X_\ell^m(\mathbf{r}) = \sum_{t,u,v}^{t+u+v=\ell} D_{t,u,v}^{\ell,m} \, x^t \, y^u \, z^v$$

$$\Phi[\rho](\mathbf{r}) = \frac{A[\rho](\mathbf{r})}{|\mathbf{r}+\mathbf{R}|} + \frac{B[\rho](\mathbf{r})}{|\mathbf{r}+\mathbf{R}|^2} + \frac{C[\rho](\mathbf{r})}{|\mathbf{r}+\mathbf{R}|^3} + \frac{D[\rho](\mathbf{r})}{|\mathbf{r}+\mathbf{R}|^4} + \dots$$

		Cauchy Test		
	$\|f(\mathbf{r})\|$	1D	2D	3D
Charge	$1/r$	$ln(R)_*$	R_{**}	$R^2{}_{**}$
Dipole	$1/r^2$	R^{-1}	$ln(R)_*$	R_{**}
Quadrupole	$1/r^3$	R^{-2}	R^{-1}	$ln(R)_*$
Octupole	$1/r^4$	R^{-3}	R^{-2}	R^{-1}

Table (1): *Convergence of the potential due to distributions of point multipoles of different order for different dimensionalities. Terms highlighted with ** are not convergent, while those with a single * are conditionally convergent. All the other terms, giving rise to negative powers of r, are absolutely convergent.*

- In the 3D case, the Coulomb sum will ***diverge*** unless the charge and dipole are *zero*. In a usual calculation, we must employ charge neutrality

$$\rho(\mathbf{r}) = \sum_A Z_A \delta(\mathbf{r} - \mathbf{r}_A) + n(\mathbf{r})$$

$$\int_{\text{cell}} d^3r \; \rho(\mathbf{r}) = 0$$

- The dipole has a value which depends on the shape of the unit cell and the crystal surface. We can *always* define the cell so as to ensure

$$\int_{\text{cell}} d^3r \; \mathbf{r}\rho(\mathbf{r}) = 0$$

- Therefore, conventionally the *leading multipole moment* of the 3D system is the quadrupole $\int_{\text{cell}} d^3r \; \mathbf{r} \otimes \mathbf{r}\rho(\mathbf{r}) \neq 0$

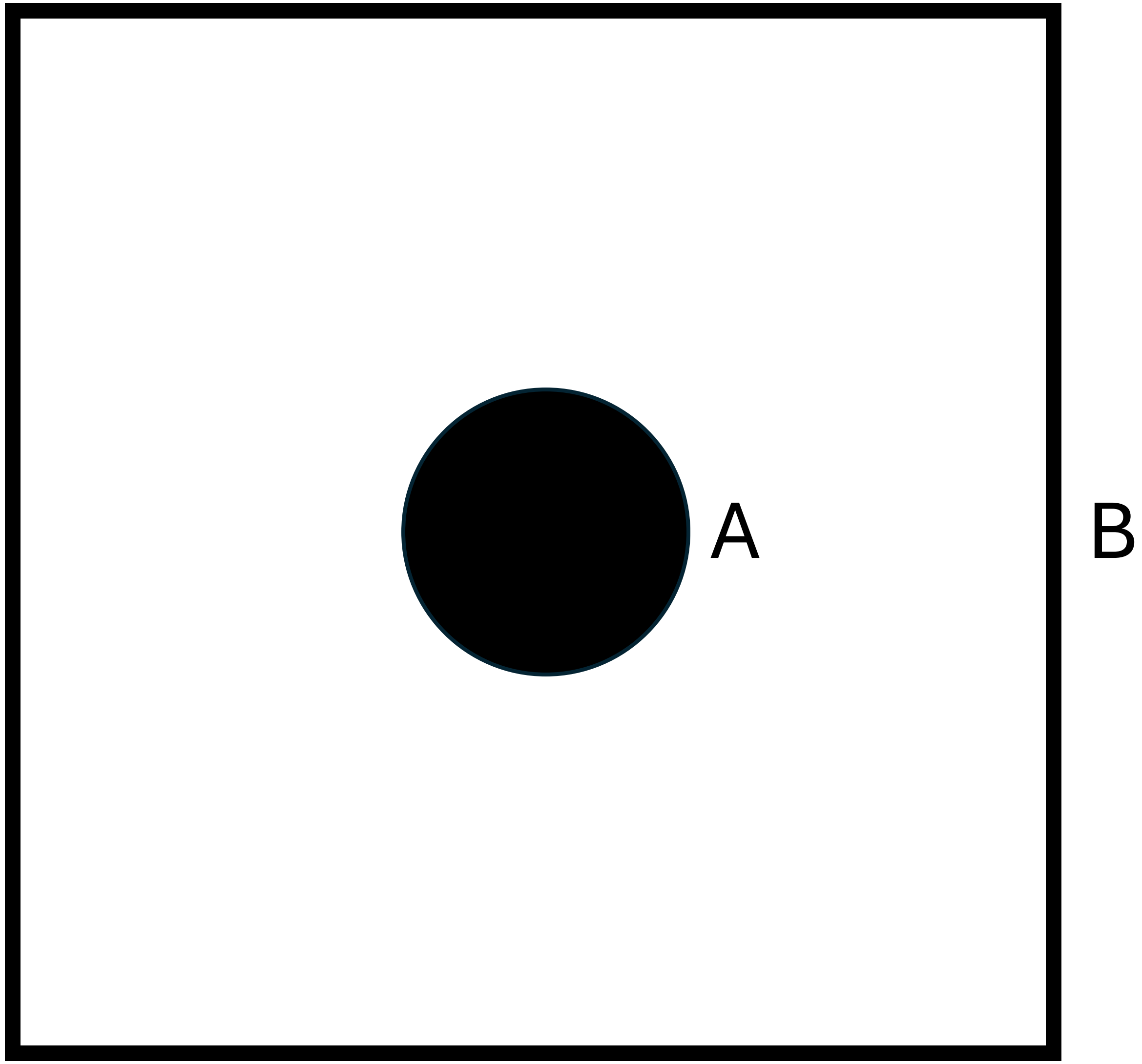
- In practice, *the 3D Coulomb* ***summation is conditionally convergent***

Ewald summation of a conditionally convergent series

3. ***Die Berechnung optischer und elektrostatischer Gitterpotentiale;***

von P. P. Ewald.

Ewald, Paul P. "Die Berechnung optischer und elektrostatischer Gitterpotentiale." *Annalen der physik* **369.3** (1921): 253-287.



$$\lim_{B\to\infty}\int_A^B dr\, r f(r)$$

- Conditionally convergent: integral may converge depending on shape of B
- Summing by squares, circles, elipses gives different results

- Convergence will depend on method of summation
- We return to

$$\Phi^{\mathrm{Coul}}[\rho]\,(\mathbf{r})$$

$$\Phi[\rho](\mathbf{r}) = \int_{\mathrm{cell}} d^3r' \sum_{\mathbf{R}} \frac{\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'+\mathbf{R}|}$$

- And introduce convergence factors by the replacement

$$\sum_{\mathbf{R}} \frac{1}{|\mathbf{x}+\mathbf{R}|} \to \sum_{\mathbf{R}} \frac{e^{-s|\mathbf{R}|^2}}{|\mathbf{x}+\mathbf{R}|} = \psi(s;\mathbf{x})$$

- Then, if it is finite, Φ will be given by

$$\Phi[\rho](\mathbf{r}) = \int_{\mathrm{cell}} d^3r'\ \rho(\mathbf{r}') \lim_{s\to 0} \psi(s;\mathbf{r}-\mathbf{r}')$$

- This is the Ewald summation procedure “*analytic continuation on the real line*”

- From the integral representation of the Gamma function

$$\psi(s;\mathbf{x}) = \sum_{\mathbf{R}} \frac{e^{-s|\mathbf{R}|^2}}{|\mathbf{x}+\mathbf{R}|} = \frac{1}{\sqrt{\pi}} \sum_{\mathbf{R}} \int_0^\infty dt\, t^{-1/2} e^{-t(\mathbf{x}+\mathbf{R})^2} e^{-s\mathbf{R}^2}$$

- The integral is *singular* at $s = 0$ for small t. Split the integral between $[0,\gamma]$ and $[\gamma,\infty]$ (choice of γ to be discussed), the 2nd integral can be evaluated analytically

$$\psi(s;\mathbf{x}) = \frac{1}{\sqrt{\pi}} \sum_{\mathbf{R}} \int_0^\gamma dt\, t^{-1/2} e^{-t(\mathbf{x}+\mathbf{R})^2} e^{-s\mathbf{R}^2} + \sum_{\mathbf{R}} \frac{\mathrm{erfc}(\sqrt{\gamma}|\mathbf{x}+\mathbf{R}|)}{|\mathbf{x}+\mathbf{R}|} e^{-s\mathbf{R}^2}$$

- Fourier expansion of the Gaussian in the reciprocal lattice provides (using also the orthogonality relation $e^{i\boldsymbol{R}\cdot\boldsymbol{G}} = 1$)

$$\sum_{\mathbf{R}} e^{-t(\mathbf{x}+\mathbf{R})^2} = \frac{\pi^{3/2}}{V t^{3/2}} \sum_{\mathbf{G}} e^{-\mathbf{G}^2/4t} e^{i\mathbf{G}\cdot\mathbf{x}}$$

- From the Fourier expansion, we obtain the integral

$$\frac{\pi}{V}\sum_{\mathbf{G}}\int_0^{\gamma} dt\ (t+s)^{-3/2}t^{-1/2}e^{-\mathbf{G}^2/4(s+t)}e^{i\mathbf{G}\cdot\mathbf{x}t/(t+s)}e^{-st/(t+s)\mathbf{x}^2}$$

- which is *singular* for $\boldsymbol{G}=\mathbf{0}$. Splitting $\boldsymbol{G}=\mathbf{0}$ term and change of variables to $t/(s+t)$ and put back in $\psi(s;\boldsymbol{x})$ gives

$$\lim_{s\to 0}\psi(s;\mathbf{x}) = \sum_{\mathbf{R}}\frac{\operatorname{erfc}(\sqrt{\gamma}|\mathbf{x}+\mathbf{R}|)}{|\mathbf{x}+\mathbf{R}|} + \frac{4\pi}{V}\sum_{\mathbf{G}\neq 0}\frac{1}{\mathbf{G}^2}e^{-\mathbf{G}^2/4\gamma}e^{i\mathbf{G}\cdot\mathbf{x}} + \lim_{s\to 0}\frac{\pi}{Vs}\int_0^{\gamma/(\gamma+s)} dt\, e^{-st\mathbf{x}^2}t^{-1/2}$$

- With the singularity isolated in the last term. Expansion of this term to first order in s provides

$$\frac{2\pi}{Vs} - \frac{\pi}{V\gamma} - \frac{2\pi}{3V}\mathbf{x}^2 + \mathcal{O}(s)$$

- Where the singularity for small s is now entirely contained in $2\pi/Vs$ (which vanishes by charge neutrality). We have calculated the “Ewald potential”!

- Return
- We get

$$\Phi[\rho](\mathbf{r}) = \int_{\text{cell}} d^3r' \; \rho(\mathbf{r}') \lim_{s\to 0} \psi(s; \mathbf{r} - \mathbf{r}')$$

$$\Phi[\rho](\mathbf{r}) = \int_{\text{cell}} d^3r' \, \rho(\mathbf{r}') \left[-\frac{\pi}{V\gamma} + \sum_{\mathbf{R}} \frac{\operatorname{erfc}(\sqrt{\gamma}|\mathbf{r} - \mathbf{r}' + \mathbf{R}|)}{|\mathbf{r} - \mathbf{r}' + \mathbf{R}|} + \frac{4\pi}{V} \sum_{\mathbf{G}\neq 0} \frac{1}{\mathbf{G}^2} e^{-\mathbf{G}^2/4\gamma} e^{i\mathbf{G}\cdot(\mathbf{r}-\mathbf{r}')} \right] - Q[\rho]$$

- With the "quadrupole" interaction

$$Q[\rho] = \frac{2\pi}{3V} \int_{\text{cell}} d^3r' \; \rho(\mathbf{r}') \mathbf{r}'^2$$

$$\Phi^{\mathrm{Ew}}[\rho]\,(\mathbf{r})$$

- A shape-dependent shift in the potential that is obtained here for macroscopic spherical samples of the 3D crystal

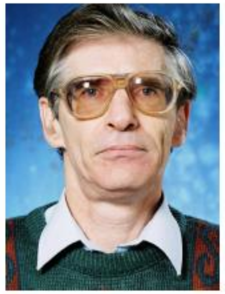
- Some comments are in order

$$\Phi[\rho](\mathbf{r}) = \int_{\text{cell}} d^3r'\, \rho(\mathbf{r}') \left[-\frac{\pi}{V\gamma} + \sum_{\mathbf{R}} \frac{\operatorname{erfc}(\sqrt{\gamma}|\mathbf{r}-\mathbf{r}'+\mathbf{R}|)}{|\mathbf{r}-\mathbf{r}'+\mathbf{R}|} + \frac{4\pi}{V} \sum_{\mathbf{G}\neq 0} \frac{1}{\mathbf{G}^2} e^{-\mathbf{G}^2/4\gamma} e^{i\mathbf{G}\cdot(\mathbf{r}-\mathbf{r}')} \right] - Q[\rho]$$

- For small γ reciprocal sum will converge quickly, but direct space sum will converge slowly (and vice versa for large γ)
- Choose γ to optimize efficiency
- Needing a zero-dipole cell is inconvenient
- Series are slowly convergent
- During the SCF process, the zero-dipole cell changes from cycle to cycle

Method in CRYSTAL

Molecular Physics, 1992, Vol. 77, No. 4, 629–665



Vic R. Saunders

Cesare Pisani

Roberto Dovesi

Carla Roetti

On the electrostatic potential in crystalline systems where the charge density is expanded in Gaussian functions

By V. R. SAUNDERS

Daresbury Laboratory, Science and Engineering Research Council, Daresbury, Warrington, WA4 4AD, UK

C. FREYRIA-FAVA, R. DOVESI, L. SALASCO and C. ROETTI

Department of Inorganic, Physical and Materials Chemistry, University of Torino, Via Giuria 5, I-10125 Torino, Italy

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$$\Phi[n_c](\mathbf{r}) = \Phi^{\mathrm{Ew}}[n_c^{\mathrm{model}}](\mathbf{r}) + \Phi^{\mathrm{Coul}}[n_c - n_c^{\mathrm{model}}](\mathbf{r}) + Q[n_c - n_c^{\mathrm{model}}]$$

$$\Phi[n_c](\mathbf{r}) = \Phi^{\text{Ew}}[n_c^{\text{model}}](\mathbf{r}) + \Phi^{\text{Coul}}[n_c - n_c^{\text{model}}](\mathbf{r}) + Q[n_c - n_c^{\text{model}}]$$

- Mulliken partition of the electron density $n(\mathbf{r}) = \sum_c^{\text{atoms}} n_c(\mathbf{r})$

$$n_c(\mathbf{r}) = \sum_{i,j} \sum_{\nu} \sum_{\mu \in c} P_{\mu \mathbf{0}, \nu \mathbf{R}_j} \chi_\mu(\mathbf{r} - \mathbf{R}_i) \chi_\nu(\mathbf{r} - \mathbf{R}_j - \mathbf{R}_i)$$

- Model density of *distributed point multipoles*

$$n_c^{\text{model}}(\mathbf{r}) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} \eta_m^\ell[n_c](\mathbf{C}_c) \delta_m^\ell(\mathbf{C}_c; \mathbf{r})$$

- $L = 2$ formally needed for absolute convergence of Φ^{Coul} term
- At present, up to $L = 6$ is implemented (default value $L = 4$)
- Higher values of L make Φ^{Coul} more short-ranged

$$n_c^{\mathrm{model}}(\mathbf{r}) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} \eta_m^\ell[n_c](\mathbf{C}_c)\delta_m^\ell(\mathbf{C}_c;\mathbf{r})$$

- δ_m^ℓ is *defined* by the property that it reproduce a given term in the Neumann-Laplace expansion of the potential

$$\Phi^{\mathrm{Coul}}[\delta_m^\ell(\mathbf{C};\mathbf{r})] = \sum_{\mathbf{R}} Z_m^\ell(\partial/\partial\mathbf{C}) \frac{1}{|\mathbf{r}+\mathbf{R}-\mathbf{C}|}$$

$$\Phi^{\mathrm{Ew}}[\delta_m^\ell(\mathbf{C};\mathbf{r})] = Z_m^\ell(\partial/\partial\mathbf{C}) \left[-\frac{\pi}{V\gamma} + \sum_{\mathbf{R}} \frac{\mathrm{erfc}(\sqrt{\gamma}|\mathbf{r}-\mathbf{C}+\mathbf{R}|)}{|\mathbf{r}-\mathbf{C}+\mathbf{R}|} + \frac{4\pi}{V} \sum_{\mathbf{G}\neq 0} \frac{1}{\mathbf{G}^2} e^{-\mathbf{G}^2/4\gamma} e^{i\mathbf{G}\cdot(\mathbf{r}-\mathbf{C})} \right]$$

- Has the advantage of rendering all long-range interactions to be represented by quickly convergent monoelectronic integrals

- Extension to 2D

Surface Science 49 (1975) 433–440
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THE ELECTROSTATIC POTENTIAL IN THE SURFACE REGION OF AN IONIC CRYSTAL

D.E. PARRY
Edward Davies Chemical Laboratories, University College of Wales, Aberystwyth, Cards., U.K.

Received 20 November 1974

- Extension to 1D

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ELSEVIER
Computer Physics Communications 84 (1994) 156–172
Computer Physics Communications

On the electrostatic potential in linear periodic polymers

V.R. Saunders [a], C. Freyria-Fava [b], R. Dovesi [b], C. Roetti [b]
[a] *Daresbury Laboratory, Science and Engineering Research Council, Daresbury, Warrington, WA4 4AD, UK*
[b] *Department of Inorganic, Physical and Materials Chemistry, University of Torino, Via Giuria 5, I-10125 Torino, Italy*

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- Atomic gradient in 3D

Doll, K., Saunders, V. R., & Harrison, N. M. (2001). Analytical Hartree–Fock gradients for periodic systems. *International Journal of Quantum Chemistry*, *82*(1), 1-13.

- Cell gradient in 3D

Doll, K., Dovesi, R., & Orlando, R. (2004). Analytical Hartree–Fock gradients with respect to the cell parameter for systems periodic in three dimensions. *Theoretical Chemistry Accounts*, *112*(5), 394-402.

- Gradients in 2D and 1D

Doll, K., Dovesi, R., & Orlando, R. (2006). Analytical Hartree–Fock gradients with respect to the cell parameter: systems periodic in one and two dimensions. *Theoretical Chemistry Accounts*, *115*(5), 354-360.

- Energy with SOC included

Desmarais, J. K., Flament, J. P., & Erba, A. (2020). Spin-orbit coupling in periodic systems with broken time-reversal symmetry: Formal and computational aspects. *Physical Review B*, *101*(23), 235142.

- Gradient with SOC included

Desmarais, J. K., Erba, A., & Flament, J. P. (2023). Structural relaxation of materials with spin-orbit coupling: Analytical forces in spin-current DFT. *Physical Review B*, *108*(13), 134108.

- Mixed field and atomic gradients

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